

Amora Florals

E-Commerce System

Entity Relationship Diagram (Crow's Foot)
and Data Dictionary

Notation	Crow's Foot logical ERD
DBMS	MySQL / MariaDB (utf8mb4), Laravel migrations
Schema source	https://github.com/shaeijmnz/amora_web
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This document describes the **implemented Laravel schema**: eight business tables (`users`, `categories`, `products`, `product_sizes`, `inventory_items`, `orders`, `order_items`, `deliveries`), their foreign keys, cardinality, and a full data dictionary. Use **Amora_Florals_ERD.pdf** (A3 landscape) as the diagram page.

1. Relationships and cardinality

Entities	Type	Cardinality	Join	On delete
USERS → ORDERS	One-to-Many	1 : N	orders.customer_id → users.id	CASCADE
ORDERS → ORDER_ITEMS	One-to-Many	1 : N	order_items.order_id → orders.id	CASCADE
ORDERS → DELIVERIES	One-to-One	1 : 0..1	deliveries.order_id → orders.id (UNIQUE)	CASCADE
PRODUCTS → PRODUCT_SIZES	One-to-Many	1 : N	product_sizes.product_id → products.id	CASCADE
PRODUCTS → ORDER_ITEMS	One-to-Many	1 : 0..N	order_items.product_id → products.id	SET NULL
PRODUCT_SIZES → ORDER_ITEMS	One-to-Many	1 : 0..N	order_items.size_id → product_sizes.id	SET NULL
CATEGORIES → INVENTORY_ITEMS	One-to-Many	1 : 0..N	inventory_items.category_id → categories.id	SET NULL
PRODUCTS → INVENTORY_ITEMS	One-to-Many	1 : 0..N	inventory_items.product_id → products.id	SET NULL

USERS → ORDERS. A customer can place many orders. Every order belongs to exactly one user.

ORDERS → ORDER_ITEMS. An order contains one or more line items. Deleting an order removes its items.

ORDERS → DELIVERIES. Each order may have at most one delivery record (created when scheduled).

PRODUCTS → PRODUCT_SIZES. An arrangement can be sold in multiple sizes, each with its own price.

PRODUCTS → ORDER_ITEMS. A catalog product can appear on many order lines. product_id is nullable for custom items.

PRODUCT_SIZES → ORDER_ITEMS. Chosen size snapshots unit price onto the line item. Nullable for custom items.

CATEGORIES → INVENTORY_ITEMS. Inventory rows optionally belong to a flower / material / add-on category.

PRODUCTS → INVENTORY_ITEMS. Optional link when a sellable product is also tracked as a stocked item.

2. Controlled vocabularies

Status columns are VARCHAR in MySQL. The application only writes these values.

Column	Allowed values
users.role	customer, admin

Column	Allowed values
categories.group	flower, material, addon
inventory_items.status	in_stock, low_stock, out_of_stock, reserved, damaged, spoiled
orders.order_type	standard, custom
orders.status	pending, confirmed, being_prepared, ready_for_delivery, dispatched, delivered, completed, cancelled, refunded
orders.payment_status	unpaid, partially_paid, paid, refunded
deliveries.status	unscheduled, scheduled, assigned, preparing_for_dispatch, dispatched, out_for_delivery, delivered, delivery_failed, rescheduled

3. Data dictionary

users

Accounts for customers and admin-dashboard staff.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
name	VARCHAR(255)	—	NO	Display name.
email	VARCHAR(255)	UK	NO	Unique login identifier.
role	VARCHAR(255)	—	NO	customer or admin. Default: customer.
phone	VARCHAR(255)	—	YES	Contact number.
email_verified_at	TIMESTAMP	—	YES	Set after OTP / email verification.
password	VARCHAR(255)	—	NO	Hashed password.
otp_code	VARCHAR(10)	—	YES	One-time passcode.
otp_expires_at	TIMESTAMP	—	YES	OTP expiry.
remember_token	VARCHAR(100)	—	YES	Laravel remember-me token.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

categories

Groups inventory items into flower, material, or add-on.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
name	VARCHAR(255)	—	NO	Category label, e.g. Fresh Flowers.
group	VARCHAR(255)	—	NO	flower material addon. Default: flower.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

products

Sellable floral arrangements in the shop catalog.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
name	VARCHAR(255)	—	NO	Arrangement name.
category	VARCHAR(255)	—	YES	Free-text shop category (NOT an FK to categories).
description	TEXT	—	YES	Product description.
note	TEXT	—	YES	Product note.
primary_image_url	VARCHAR(255)	—	YES	Main catalog image.
gallery_image_urls	JSON	—	YES	Additional gallery images.
images	JSON	—	YES	Alternate image payload.
preparation_time_minutes	INT UNSIGNED	—	NO	Estimated prep time. Default: 60.
is_available	BOOLEAN	—	NO	Whether the product can be ordered.
is_featured	BOOLEAN	—	NO	Highlighted on the storefront.
is_stem	BOOLEAN	—	NO	Single-stem item vs bouquet.
is_customisable	BOOLEAN	—	NO	Customer may pick colors / extras.
customisation_items	JSON	—	YES	Allowed customisation options.
rating	DECIMAL(3,1)	—	NO	Display rating. Default: 4.5.
reviews_count	INT UNSIGNED	—	NO	Review count. Default: 0.
archived_at	TIMESTAMP	—	YES	Soft-delete timestamp.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

product_sizes

Size/price variants (Small, Medium, Large).

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
product_id	BIGINT UNSIGNED	FK	NO	Parent product. CASCADE on delete.
label	VARCHAR(255)	—	NO	Size label.
price	DECIMAL(10,2)	—	NO	Unit price for this size.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

inventory_items

Physical stock of flowers, materials, and add-ons.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
product_id	BIGINT UNSIGNED	FK	YES	Optional catalog product. SET NULL on delete.
category_id	BIGINT UNSIGNED	FK	YES	Inventory category. SET NULL on delete.
name	VARCHAR(255)	—	NO	Stock item name.
image_url	VARCHAR(255)	—	YES	Photo of the stock item.
unit	VARCHAR(255)	—	NO	Unit of measure. Default: pcs.
quantity_on_hand	INT	—	NO	Current stock quantity. Default: 0.
min_stock_level	INT	—	NO	Low-stock threshold. Default: 5.
status	VARCHAR(255)	—	NO	Stock status. Default: in_stock.
expiration_date	DATE	—	YES	Freshness date for perishable flowers.
archived_at	TIMESTAMP	—	YES	Soft-delete timestamp.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

orders

Customer checkout records, gift recipient, and PayMongo payment fields.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
order_number	VARCHAR(255)	UK	NO	Human-readable order code.
customer_id	BIGINT UNSIGNED	FK	NO	Ordering user. CASCADE on delete.
order_type	VARCHAR(255)	—	NO	standard or custom. Default: standard.
status	VARCHAR(255)	—	NO	Fulfillment state. Default: pending.
payment_status	VARCHAR(255)	—	NO	Payment state. Default: unpaid.
payment_method	VARCHAR(255)	—	YES	GCash, card, cash, etc.
paymongo_checkout_id	VARCHAR(255)	—	YES	PayMongo Checkout Session id.
paymongo_payment_id	VARCHAR(255)	—	YES	PayMongo payment id after success.
paid_at	TIMESTAMP	—	YES	When payment was confirmed.
subtotal	DECIMAL(10,2)	—	NO	Sum of line items. Default: 0.
delivery_fee	DECIMAL(10,2)	—	NO	Delivery charge. Default: 0.
discount	DECIMAL(10,2)	—	NO	Discount amount. Default: 0.
total	DECIMAL(10,2)	—	NO	Amount due / paid. Default: 0.
recipient_name	VARCHAR(255)	—	YES	Gift recipient (may differ from customer).
recipient_contact	VARCHAR(255)	—	YES	Recipient phone number.
delivery_address	TEXT	—	YES	Free-text delivery address snapshot.
delivery_notes	TEXT	—	YES	Special delivery instructions.
admin_notes	TEXT	—	YES	Internal staff notes.
created_at	TIMESTAMP	—	YES	Order placed at.
updated_at	TIMESTAMP	—	YES	Last update timestamp.

order_items

Line items inside an order. Unit price is snapshotted at checkout.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
order_id	BIGINT UNSIGNED	FK	NO	Parent order. CASCADE on delete.
product_id	BIGINT UNSIGNED	FK	YES	Catalog product. SET NULL on delete.
size_id	BIGINT UNSIGNED	FK	YES	Chosen size. SET NULL on delete.
quantity	INT UNSIGNED	—	NO	Quantity ordered. Default: 1.
unit_price	DECIMAL(10,2)	—	NO	Price per unit at time of order.
flower_color	VARCHAR(255)	—	YES	Requested flower color.
personalized_message	TEXT	—	YES	Card message for the recipient.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

deliveries

Delivery leg of an order, including rider assignment and proof of delivery.

Attribute	Type	Key	Null	Description
id	BIGINT UNSIGNED	PK	NO	Surrogate primary key.
order_id	BIGINT UNSIGNED	FK	NO	Unique FK — one delivery per order. CASCADE.
status	VARCHAR(255)	—	NO	Delivery state. Default: unscheduled.
assigned_rider	VARCHAR(255)	—	YES	Rider name / identifier (not an FK).
scheduled_date	DATE	—	YES	Planned delivery date.
scheduled_time	VARCHAR(255)	—	YES	Planned delivery window.
delivery_instructions	TEXT	—	YES	Gate codes, landmarks, etc.
failed_reason	VARCHAR(255)	—	YES	Reason for the latest failed attempt.
proof_of_delivery_url	VARCHAR(255)	—	YES	Photo / signature URL.
attempts	JSON	—	YES	Array of prior delivery attempts.
created_at	TIMESTAMP	—	YES	Created at.
updated_at	TIMESTAMP	—	YES	Updated at.

4. Notes for defense

products.category is not an FK to categories. Catalog category is free text. The categories table is used by inventory items only.

deliveries.attempts is JSON, not a separate delivery_attempts table.

assigned_rider is a string, not a foreign key to users.

Laravel framework tables (sessions, cache, jobs, personal_access_tokens) are omitted from the ERD because they are not business entities.